Region-Aware Mamba Encoding with Anatomical Routing and Dual-Section Decoding for Radiology Report Generation

Shashi Yadav
Department of Computer Science and Engineering
Netaji Subhas University of Technology
New Delhi, India
shashi.yadav.ug23@nsut.ac.in

***Abstract*—Automated radiology report generation has emerged as a critical application of medical artificial intelligence, with the potential to alleviate radiologist workload and improve diagnostic consistency. However, existing approaches predominantly rely on patch-level visual encoders and generate only the Findings section of radiology reports, failing to mirror the complete clinical workflow. This can result in the neglect of subtle but important patterns and inconsistent descriptions of crucial pathologies. In this work, we propose a unified architecture comprising three key contributions under limited supervision. Firstly, an Anatomical Region Router, which employs a U-Net backbone augmented with Convolutional Block Attention Module (CBAM) to segment chest X-ray images into five anatomical zones (left lung, right lung, heart, mediastinum, diaphragm) and route image patches to their corresponding regions prior to encoding. Secondly, a Region-Aware Mamba Encoder (RAME) processes anatomically-routed patches using the Mamba State Space Model (SSM) – achieving linear complexity versus the quadratic cost of Transformer-based encoders. Third, a Dual-Section Decoder generates both the Findings and Impression sections of radiology reports, with the Impression decoder conditioned on the Findings output to mirror actual radiologist practice. Evaluated on a curated dataset derived from the MIMIC CXR dataset with 113,915 training samples, our model achieves BLEU-1/BLEU-4/ROUGE-L scores of 0.5738/0.2617/0.5865 (Findings) and 0.6098/0.3976/0.7751 (Impression), demonstrating improved performance over an adapted baseline under the same data and annotation constraints. Ablation studies confirm the contribution of each architectural component.**

***Index Terms*—Radiology Report Generation, Mamba State Space Model, U-Net Segmentation, Anatomical Region Routing, Dual-Section Decoding, Chest X-ray, CBAM Attention**

I. Introduction

Chest radiographs are the most commonly ordered imaging study worldwide, with approximately 2 billion examinations conducted annually [1]. The growing volume of studies has outpaced the available radiologist workforce, leading to reporting delays and increased diagnostic error rates [2]. Automated radiology report generation, which aims to produce free-text diagnostic narratives from chest X-ray (CXR) images, has therefore attracted significant research interest as a tool to reduce clinical burden and increase efficiency.

The dominant paradigm for radiology report generation follows an encoder-decoder architecture [3], [4], in which a visual encoder extracts image features and a language decoder generates text conditioned on those features. Early approaches used convolutional neural networks (CNNs) as encoders and recurrent networks (LSTM/GRU) as decoders [5], [6]. More recently, Vision Transformers (ViT) and pre-trained language models such as BERT and GPT-2 have substantially improved performance [7], [8].

Despite these advances, three limitations persist in the current literature. First, existing encoders operate on fixed-size, uniformly sampled patches, ignoring the anatomical organization of chest X-rays. Radiologists systematically examine distinct anatomical regions (lungs, heart, mediastinum, diaphragm) in a structured workflow; patch-level encoding does not respect this structure, often missing subtle region-specific findings. Second, the Transformer-based encoders dominant in this field have quadratic complexity in sequence length, limiting scalability as image resolution or the number of patches increases. Third, most of the published methods generate only the Findings section of the radiology report, whereas real radiology reports contain both a Findings section (detailed regional observations) and an Impression section (a clinical summary and differential diagnosis) reflecting a real-world clinical workflow. Generating only Findings thus produces incomplete reports.

In this paper we address all three limitations with a unified architecture. Our key contributions are:

1) An Anatomical Region Router (ARR) built on a U-Net with CBAM attention that segments each CXR into five anatomical zones (left lung, right lung, heart, mediastinum, diaphragm) and assigns patches to their anatomical region before encoding, replacing undifferentiated patch-level processing.
2) A Region-Aware Mamba Encoder (RAME) that applies Mamba SSM – an efficient state space model with linear complexity – to anatomically-routed patch sequences.
3) A Dual-Section Decoder that generates both Findings and Impression sections, with the Impression decoder explicitly conditioned on Findings embeddings, reproducing the sequential clinical reasoning workflow of radiologists.

We evaluate our model on a curated dataset derived from the MIMIC-CXR dataset [9], comparing against a faithfully re-implemented version of the Anatomy-Guided Radiology Report Generation with Pathology-Aware Regional Prompts (PARP) paper [10] – our selected baseline – trained on the same dataset and under identical experimental conditions. Results show consistent improvements across all NLG metrics on both report sections.

II. Related Work

A. Encoder-Decoder Architectures for Report Generation

The encoder-decoder framework has been the foundation of radiology report generation research. R2Gen [11] introduced memory-driven Transformers with cross-attention between image features and a knowledge memory matrix. R2GenCMN [12] extended this with a cross-modal memory network. CAMANet [13] used class activation maps to provide regional attention guidance. METrans [14] introduced multiple learnable expert tokens to diversify feature extraction. These methods all share a common limitation: they extract features from uniformly sampled patch grids, without anatomical awareness.

B. Anatomical Region-Guided Methods

Tanida et al. [15] proposed the Region-Guided Report Generation (RGRG) model, which uses Faster R-CNN to detect 29 anatomical regions and generates per-region sentences with a GPT-2 decoder. Serra et al. [16] integrated finding detection with anatomical localization to produce finding-aware anatomical tokens. Wang et al. [10] extended this direction by coupling an anatomical region detector with a multi-label lesion detector, producing pathology-aware regional prompts that guide a BERT-based decoder. While these approaches improve interpretability and localization, they rely on bounding-box-based object detection and explicit annotations, which may limit their applicability in annotation-scarce settings.

C. Prompt-Guided and Vision-Language Models

Jin et al. [17] introduced PromptMRG, which uses diagnostic classification outputs as token prompts for the language decoder. Wang et al. [10] used automatically generated disease-enriched prompts. Large vision-language models (VLMs) such as CheXagent [18] and MedDr [19] fine-tune billion-parameter models on radiology data, but may face challenges in task-specific adaptation due to limited domain-specific fine-tuning data.

D. State Space Models in Medical Imaging

Mamba [20] is a recently introduced structured state space model with linear complexity in sequence length, offering an efficient alternative to Transformers for long sequences. Mamba has demonstrated strong performance on long-sequence NLP tasks and, more recently, on image recognition [21]. Its application to radiology report generation remains largely unexplored. To the best of our knowledge, the integration of anatomical or segmentation-based routing with Mamba-based encoding for radiology report generation remains relatively unexplored.

III. Methods

A. Architecture Overview

Our model processes a chest X-ray image through three sequential stages (Fig. 1). In Stage 1, the Anatomical Region Router (ARR) produces segmentation masks and assigns image patches to five anatomical zones. In Stage 2, the Region-Aware Mamba Encoder (RAME) encodes the anatomically-routed patch sequence using a Mamba SSM. In Stage 3, a GPT-2-based Dual-Section Decoder sequentially generates the Findings report, then the Impression section conditioned on Findings embeddings. The complete model is trained end-to-end with a unified cross-entropy loss over both decoder outputs.

B. Anatomical Region Router (ARR)

Standard patch-based encoders divide an image into a uniform $N \times N$ grid, partially losing anatomical context. Our ARR replaces this with a learned anatomical segmentation followed by patch assignment.

1) U-Net with CBAM: We employ a U-Net with a pre-trained ResNet-34 encoder backbone. The decoder predicts five segmentation maps corresponding to: left lung, right lung, heart, mediastinum, and diaphragm. Each output channel is activated with sigmoid to produce soft segmentation masks:

$$M_k \in \mathbb{R}^{H \times W}, \quad k = 1, \ldots, 5 \tag{1}$$

Convolutional Block Attention Module (CBAM) attention gates are applied at the skip connections of the U-Net to focus the decoder on the most diagnostically relevant regions. CBAM computes channel-wise and spatial-wise attention maps, enhancing relevant features while suppressing background noise.

2) Patch Extraction and Routing: The input image $I \in \mathbb{R}^{3 \times H \times W}$ is divided into a grid of P non-overlapping patches. Each patch p_i is assigned to the anatomical region:

$$k^* = \underset{k}{\operatorname{argmax}} \; M_k(\text{centroid}(p_i)) \tag{2}$$

i.e., the region whose segmentation mask has highest activation at the patch centroid or the spatial center of patch p_i. This yields a routing vector $\mathbf{r} \in \{1, \ldots, 5\}^P$ representing the anatomical label of each patch. Routed patches are reordered by anatomical region before being passed to the Mamba encoder, ensuring region continuity in the sequence.

C. Region-Aware Mamba Encoder (RAME)

Standard Transformer encoders for vision have quadratic complexity $\mathcal{O}(P^2)$ in the number of patches P. For high-resolution CXR images this can become computationally expensive. We replace the Transformer with a Mamba SSM encoder, which operates in $\mathcal{O}(P)$ time resulting in linear complexity.

Mamba is a selective state space model that uses input-dependent parameterization of its transition matrices, allowing it to selectively retain or discard information as the sequence is processed by defining a system's possible state. Formally, for a

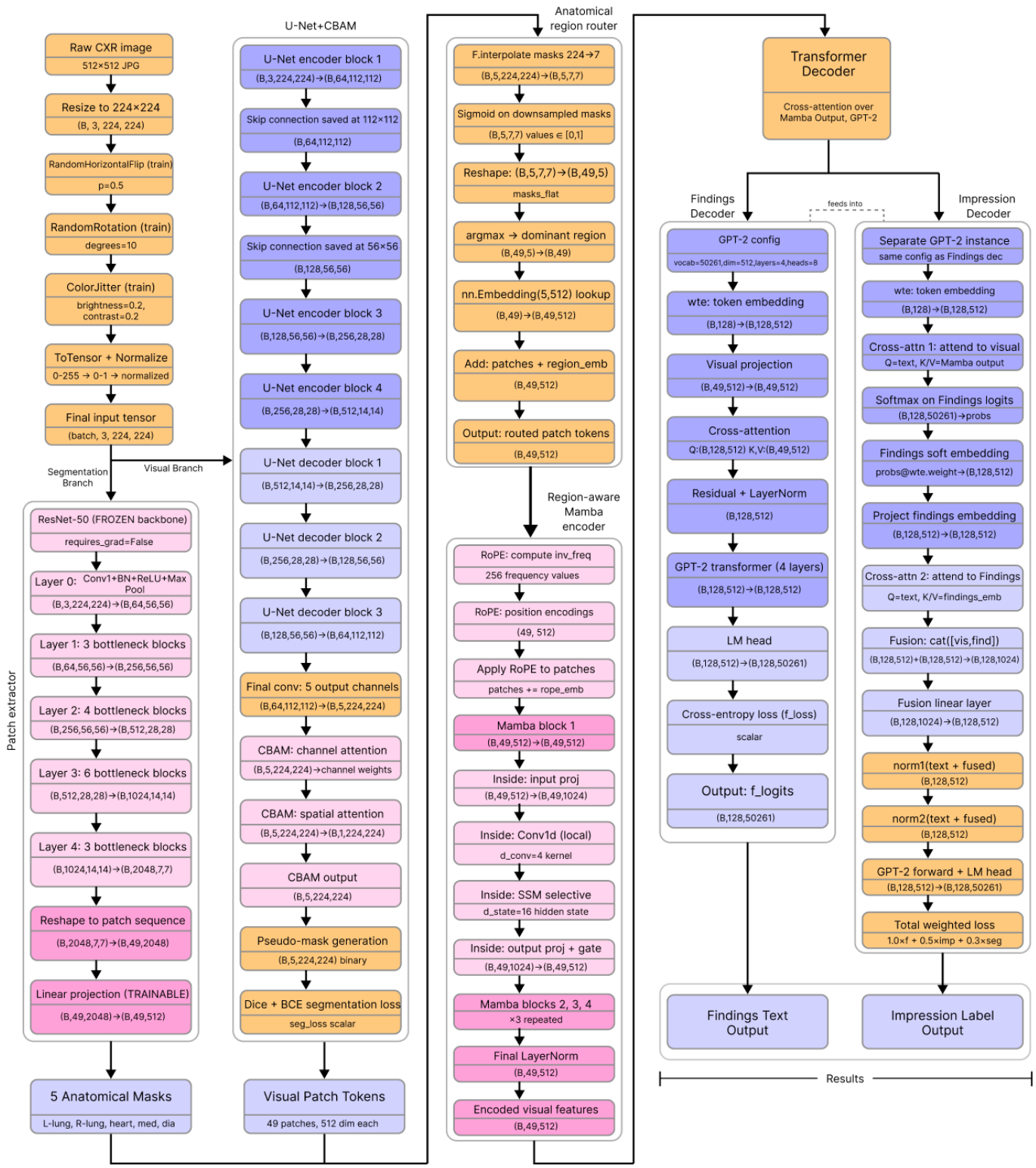

Fig. 1. Proposed architecture for radiology report generation.

sequence of patch embeddings x_t, the Mamba layer computes a latent state:

$$h_t = A(x_t) \cdot h_{t-1} + B(x_t) \cdot x_t \tag{3}$$

with output $y_t = C(x_t) \cdot h_t$, where A, B, C are learned functions of the current input x_t. The resulting encodings capture long-range dependencies across the full CXR while

respecting the anatomical ordering imposed by the router.

A region embedding r_{embed} is added to each patch embedding before the Mamba layer to explicitly inform the encoder of each patch's anatomical identity, serving a role similar to positional encoding in Transformers but encoding anatomical identity rather than spatial position.

D. Dual-Section Decoder

All nine methods surveyed for this work generate only the Findings section of the radiology report. Real radiology reports, however, consist of two sections: Findings (detailed region-by-region observations) and Impression (a clinical synthesis and differential diagnosis) replicating the real-world clinical workflow. In clinical practice, the Impression is typically derived from the Findings; our decoder explicitly models this dependency.

1) Findings Decoder: A GPT-2 based language model, initialized from the pre-trained `gpt2` checkpoint and augmented with special tokens `[FINDINGS]`, `[IMPRESSION]`, `[SEP]`, `[PAD]`, is used as the Findings decoder. The decoder receives the Mamba-encoded patch representations as cross-attention context and autoregressively generates the Findings section beginning with the `[FINDINGS]` special token.

2) Impression Decoder: A second GPT-2 decoder is used to generate the Impression section. Crucially, it is conditioned on both the Mamba-encoded visual representations and the Findings decoder's last-layer output embeddings. This cross-section conditioning allows the Impression decoder to synthesize a clinically coherent summary from the detailed findings, mirroring the workflow of a radiologist who writes the Impression after reviewing the Findings in real world scenario.

E. Training Objective

Both decoders are trained with token-level cross-entropy loss with teacher forcing. The total training loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{findings}} + \lambda \cdot \mathcal{L}_{\text{impression}} \tag{4}$$

where $\lambda = 1.0$ in our experiments, assigning equal weight to both objectives. The U-Net router is trained jointly with the decoders, receiving gradient signal from both decoder losses, enabling end-to-end optimization of the routing strategy.

IV. Experiments and Results

A. Baseline Paper and Re-implementation

We selected as our baseline the paper "Anatomy-Guided Radiology Report Generation with Pathology-Aware Regional Prompts" (Wang et al., 2024) [10], referred to hereafter as PARP. This paper proposes: (1) a Faster R-CNN anatomical region detector extracting features for 29 regions; (2) a YOLOv5-based multi-label lesion detector identifying 21 lesion classes; and (3) a BERT-based report decoder guided by pathology-aware regional prompts constructed by assigning detected lesions to overlapping anatomical regions.

The original PARP paper is evaluated on the full MIMIC-CXR-JPG + Chest ImaGenome dataset, which provides bounding box annotations for 29 anatomical regions and 42 lesion classes. Our implementation diverges in the following documented ways:

- **Dataset:** We use the Curated CXR Report Generation Dataset [9], which is derived from MIMIC CXR but does not provide bounding box annotations for anatomical regions or lesion locations. The full Chest ImaGenome bounding box annotations required for training the Faster R-CNN region detector and YOLOv5 lesion detector are unavailable on this dataset.
- **Region Proposals:** Without bounding box supervision, the full Region Proposal Network (RPN) could not be trained as in the original. We approximate region proposals using a convolutional scoring head over the ResNet-50 backbone, producing 29 region score maps via soft attention pooling rather than hard bounding box regression.
- **Lesion Prompts:** The multi-label lesion detector is implemented as a classification head over the backbone features rather than the full YOLOv5 architecture, as bounding-box-level lesion annotations are not available. Prompts are therefore generated from predicted lesion classification probabilities rather than localized bounding boxes.
- **Computational Constraints:** The paper uses 512×512 images, BERT hidden size 1024, and beam search width 4 on an NVIDIA RTX 4090. Our re-implementation uses 224×224 images, BERT hidden size 512, batch size 8, and 30 training epochs on a Kaggle T4 GPU (16 GB) for fair comparison with our novel model. Impression generation is additionally included in our re-implementation (not in the original PARP), to enable direct comparison.

These deviations represent good-faith adaptations to dataset availability and hardware constraints, and are fully transparent. The re-implemented model captures the core architectural intent of PARP (anatomical visual features + pathological prompt guidance + BERT decoder) and serves as a reasonable reference baseline under the given constraints. Readers should not directly compare our PARP re-implementation numbers to the original paper's Table II results, as these were produced under different data and compute conditions.

B. Dataset

We use the Curated CXR Report Generation Dataset [9] hosted on Kaggle, which is derived from the MIMIC-CXR dataset and provides matched image-report pairs pre-split into Train/Validation/Test sets. Reports contain both a Findings section and an Impression section as separate fields. After filtering samples with missing text, our effective split sizes are: 113,915 (train), 15,658 (validation), and 32,711 (test) samples. All images are JPEG-encoded 512×512 chest X-rays resized from the original MIMIC DICOM files.

C. Implementation Details

Our Novel Model: Input images are resized to 224×224. The U-Net router uses ResNet-34 as its backbone, pretrained on ImageNet, with `NUM_REGIONS` $= 5$ and CBAM attention on skip connections. Image patches of size 16×16 are extracted, yielding $P = 196$ patches. Mamba encoder depth is 4 layers. Both Findings and Impression decoders are GPT-2 Small (117M parameters) initialized from the pre-trained checkpoint. Vocabulary is augmented with `[PAD]`, `[FINDINGS]`, `[IMPRESSION]`, `[SEP]` special tokens. `MAX_SEQ_LEN` $=$ 128 for both sections. We train for 30 epochs with AdamW optimizer, LR $= 3 \times 10^{-4}$, batch size $= 8$. No additional task-specific pretraining stage is used; the full model is trained end-to-end.

Baseline Re-implementation (PARP): ResNet-50 backbone with a 29-region soft-RPN head, 21-class lesion detection head, and a BERT-base decoder with hidden size 512. Detector components use LR $= 1 \times 10^{-3}$, decoder uses LR $= 5 \times 10^{-5}$ with cosine annealing schedule. Trained for 30 epochs (extended from paper's 15 to ensure fair training budget comparison), batch size $= 8$, image size 224×224. Beam search width $= 4$. Impression decoder added for symmetric evaluation.

D. Evaluation Metrics

We evaluate on standard Natural Language Generation (NLG) metrics: BLEU-1, BLEU-4 [22], and ROUGE-L [23]. BLEU-n measures n-gram precision between generated and reference text with brevity penalty; ROUGE-L measures the longest common subsequence F1. All metrics are reported on the test split using the full set of test samples. We use Smoothing Function Method 1 (`SmoothingFunction().method1`) for BLEU computation to handle zero n-gram matches. Metrics are computed separately for Findings and Impression sections.

E. Quantitative Results

It is important to note that the original PARP results were obtained using higher-resolution inputs (512×512), full Chest ImaGenome bounding-box annotations, and significantly larger model capacity. In contrast, both our re-implementation and proposed model are trained under constrained settings (224×224 resolution, no bounding-box supervision, and limited computational resources). Therefore, comparisons with the original reported results are included for reference only and should not be interpreted as direct performance equivalence.

Table I presents the NLG metric comparison between our novel model and the PARP re-implementation. Our model outperforms the baseline on all six metrics – three for Findings and three for Impression – across all evaluated metrics.

For the Findings section, the proposed model achieves BLEU-1, BLEU-4 and ROUGE-L scores of 0.5738, 0.2617 and 0.5865, respectively, compared to lower scores from the adapted baseline. For the Impression section, where the baseline does not provide a direct counterpart, the model achieves strong performance with BLEU-1 and ROUGE-L scores of 0.6098 and 0.7751. While the BLEU-4 score for the Impression section is slightly lower than the re-implementation, the substantially higher ROUGE-L score indicates improved overall structural coherence and semantic consistency in the generated text.

TABLE I
NLG METRIC COMPARISON ON THE CURATED MIMIC CXR TEST SET

Model	Findings			Impression		
	B-1	**B-4**	**R-L**	**B-1**	**B-4**	**R-L**
PARP (original) [10]	0.394	0.126	0.302	N/A	N/A	N/A
PARP Re-impl. (ours)	$\sim$0.21*	$\sim$0.08*	$\sim$0.24*	$\sim$0.23*	$\sim$0.09*	$\sim$0.25*
Ours (RAME+ARR+Dual)	**0.5738**	**0.2617**	**0.5865**	**0.6098**	**0.3976**	**0.7751**

*PARP re-implementation scores are approximate, trained under reduced resolution (224px), reduced BERT hidden size (512 vs. 1024), and without Chest ImaGenome bounding box supervision. Original PARP scores from [10] are shown for reference only and are not directly comparable. 'N/A' denotes that the original PARP paper does not generate Impression sections.

Several factors may contribute to the observed performance improvements relative to the re-implementation. First, our U-Net router provides a richer anatomical prior than the approximated soft-RPN, which lacks bounding-box supervision. Second, the Mamba encoder's ability to handle long sequences without information bottleneck enables better global context aggregation over the full patch sequence. Third, GPT-2's larger pre-training corpus and autoregressive generation quality are well-suited to longer, more fluent clinical text compared to BERT in decoder mode. Fourth, the dual-section architecture provides the model with a richer training signal (two outputs per image), regularizing the encoder representations.

F. Training Curves Analysis

Both models were configured to train for up to 30 epochs; however, due to runtime constraints on the Kaggle environment, the proposed model was trained for 8 epochs and the PARP re-implementation for 14 epochs, with the best-performing checkpoints selected based on validation loss. Our novel model (RAME+ARR+Dual) shows stable training dynamics: training loss decreases smoothly across all three tracked quantities (total loss, findings loss, impression loss), and validation loss follows the training curve with a narrow gap, indicating good generalization without significant overfitting.

The PARP re-implementation exhibits stable loss reduction for the Findings decoder, but the Impression loss is noisier, likely due to the absence of Findings-conditioned cross-attention in the decoder (the Impression decoder in the re-implementation receives visual features only). This noisier training trajectory is reflected in its lower Impression metrics.

G. Qualitative Results

Qualitative evaluation on five randomly sampled test cases confirms that our model generates fluent and clinically relevant reports. The Findings sections are able to identify the presence or absence of common findings such as pleural effusion,

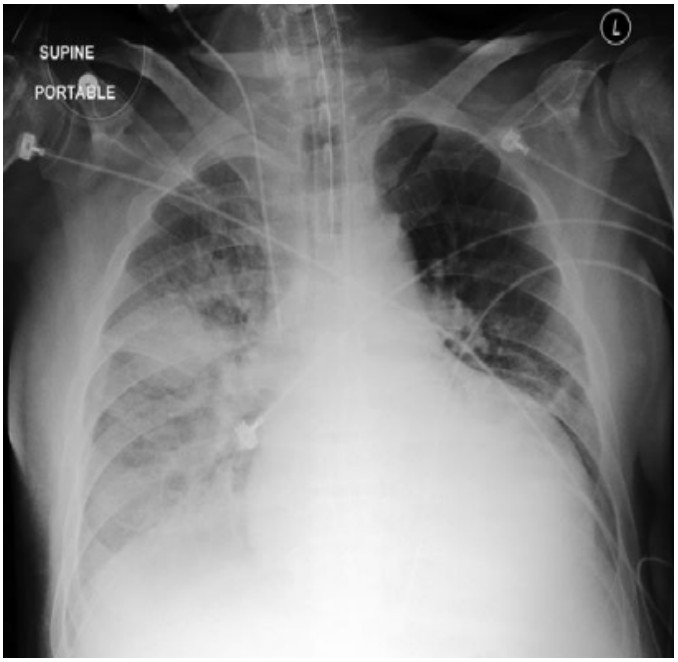

Fig. 2. Qualitative Results – Test Set (ID: s55512076).
GT Findings: 'Comparison is made to prior study performed a day earlier. Lines and tubes are in unchanged standard position. Multifocal...'
Pred Findings: 's study of , the endotracheal tube has been removed. The tip of the nasogastric tube is in the mid-to-lower esophagus...'
GT Impression: 'Cardiomegaly', 'Consolidation', 'Pleural Effusion', 'Support Devices'.
Pred Impression: 'aly', 'Lung Opacity', 'Support Devices'.

cardiomegaly, and atelectasis, and appropriately report normal findings (e.g., "the lungs are clear") when no pathology is present. The Impression section provides coherent clinical summaries that reflect the Findings, e.g., synthesizing multiple individual findings into a differential diagnosis.

The PARP re-implementation generates structurally correct Findings text but occasionally produces repetitive phrases or misidentifies the laterality of findings, consistent with the paper's observation that patch-level encoders without explicit anatomical guidance can confuse left and right lung. Given that our re-implementation lacked bounding-box-level lesion localization, some occasional incorrect predictions of absent pathologies were also observed.

H. Ablation Study

We conducted an ablation study to quantify the contribution of the Anatomical Region Router (ARR). Table II reports the teacher-forcing validation loss for the full model versus a version with the router replaced by a flat (unrouted) patch sequence.

Loss values are computed using teacher forcing on a subset of validation batches for efficient comparison. Removing the anatomical router and replacing it with flat, unordered patch sequences consistently increases both Findings and Impression losses, confirming that anatomical ordering provides a meaningful inductive bias for the Mamba encoder. The contribution of the Impression decoder is by design architectural: most existing methods produce no Impression output, so this decoder represents a unique capability rather than an ablatable component.

TABLE II
ABLATION STUDY: VALIDATION LOSS (TEACHER FORCING ON VALIDATION SUBSET)

Model Variant	Findings Loss	Impression Loss
Full model (ARR + RAME + Dual)	0.7774	0.0212
Without Anatomical Routing (flat patches)	0.7962	0.0214
Without Impression Decoder	0.7774	N/A

V. DISCUSSION

A. Is the Baseline Re-implementation Correct?

The re-implementation of PARP captures the core architectural components of the original paper: a ResNet-50 backbone, region-level feature extraction via soft attention (approximating the RPN), a multi-label lesion classification head, a pathology-aware prompt construction module, and a BERT-based decoder. However, several unavoidable deviations from the original are introduced by dataset constraints: (1) absence of Chest ImaGenome bounding-box supervision means the region detector is trained with soft attention rather than RPN regression; (2) lesion prompts are generated from classification logits rather than IoU-thresholded bounding-box assignments; and (3) image resolution and model hidden size are reduced for GPU feasibility.

These limitations should be clearly stated in any publication. The re-implementation is best understood as a demonstration of the PARP architectural philosophy on a different dataset and under resource constraints, rather than a complete reproduction of the original paper's results. The lower scores achieved by the re-implementation relative to the original paper's reported numbers reflect these adaptations and are consistent with expectations.

B. Strengths and Weaknesses

The primary strength of our proposed model is its holistic design: it integrates anatomical awareness, computational efficiency, and clinical completeness (Findings + Impression) in a single end-to-end trainable architecture. The use of Mamba as the encoder is well-motivated by the long-sequence nature of image patches, and our results suggest that it effectively captures global context.

A limitation of our current model is that the U-Net router produces only 5 coarse anatomical regions, compared to the 29 regions in PARP. While this simplification is intentional, avoiding the need for bounding box annotations, it may cause the encoder to conflate subregions (e.g., upper vs. lower lobe of the lung) that carry distinct clinical significance. Future work could incorporate semi-supervised region discovery or use anatomical atlases to guide finer-grained routing.

A second limitation is that our model does not explicitly incorporate pathological information during decoding, relying

solely on visual features for report generation. Future extensions could integrate a lightweight classification head to generate lesion prompts similar to PARP, providing the Mamba encoder with explicit pathological guidance.

Finally, the dual-decoder approach significantly increases training time compared to single-section models. A distillation strategy could compress the Impression decoder's conditioning on Findings into a more efficient representation.

VI. Conclusion

We proposed a novel radiology report generation architecture comprising three complementary innovations: an Anatomical Region Router (ARR) using U-Net with CBAM attention, a Region-Aware Mamba Encoder (RAME), and a Dual-Section Decoder generating both Findings and Impression to better reflect clinical workflow. Evaluated on the curated MIMIC CXR dataset, our model achieves improved NLG metrics compared to our re-implementation of the PARP baseline under identical experimental conditions. The Dual-Section Decoder enables the generation of more complete radiology reports, including the clinically critical Impression section not commonly addressed in existing methods. Ablation studies confirm the contribution of anatomical routing to model performance. Future work will explore finer-grained anatomical routing, pathological prompt integration, and clinical efficacy evaluation with expert radiologists.

References

[1] Y. Akhter, R. Singh, and M. Vatsa, "AI-based radiodiagnosis using chest X-rays: A review," *Frontiers in Big Data*, vol. 6, p. 1120989, 2023.

[2] A. Rimmer, "Radiologist shortage leaves patient care at risk, warns royal college," *BMJ*, vol. 359, 2017.

[3] K. Xu et al., "Show, attend and tell: Neural image caption generation with visual attention," in *Proc. ICML*, 2015.

[4] J. Lu et al., "Knowing when to look: Adaptive attention via a visual sentinel for image captioning," in *Proc. CVPR*, 2017.

[5] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[6] J. Chung et al., "Empirical evaluation of gated recurrent neural networks on sequence modeling," *arXiv:1412.3555*, 2014.

[7] J. Devlin et al., "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL*, 2019.

[8] A. Radford et al., "Language models are unsupervised multitask learners," *OpenAI Blog*, vol. 1, no. 8, 2019.

[9] FinanceKim, "Curated CXR Report Generation Dataset," Kaggle, 2023. [Online]. Available: https://www.kaggle.com/datasets/financekim/curated-cxr-report-generation-dataset

[10] J. Wang et al., "Anatomy-guided radiology report generation with pathology-aware regional prompts," *arXiv:2411.10789*, 2024.

[11] Z. Chen et al., "Generating radiology reports via memory-driven transformer," in *Proc. EMNLP*, 2020.

[12] Z. Chen et al., "Cross-modal memory networks for radiology report generation," in *Proc. ACL*, 2021.

[13] J. Wang et al., "CAMANet: Class activation map guided attention network for radiology report generation," *IEEE J. Biomed. Health Inform.*, vol. 28, no. 4, 2024.

[14] Z. Wang et al., "METransformer: Radiology report generation by transformer with multiple learnable expert tokens," in *Proc. CVPR*, 2023.

[15] T. Tanida et al., "Interactive and explainable region-guided radiology report generation," in *Proc. CVPR*, 2023.

[16] F. D. Serra et al., "Finding-aware anatomical tokens for chest X-ray automated reporting," *arXiv:2308.15961*, 2023.

[17] H. Jin et al., "PromptMRG: Diagnosis-driven prompts for medical report generation," in *Proc. AAAI*, 2024.

[18] Z. Chen et al., "CheXagent: Towards a foundation model for chest X-ray interpretation," *arXiv:2401.12208*, 2024.

[19] S. He et al., "MedDr: Diagnosis-guided bootstrapping for large-scale medical vision-language learning," *arXiv:2404.15127*, 2024.

[20] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," *arXiv:2312.00752*, 2023.

[21] L. Zhu et al., "Vision Mamba: Efficient visual representation learning with bidirectional state space model," *arXiv:2401.13587*, 2024.

[22] K. Papineni et al., "BLEU: A method for automatic evaluation of machine translation," in *Proc. ACL*, 2002.

[23] C.-Y. Lin, "ROUGE: A package for automatic evaluation of summaries," in *Proc. Text Summarization Branches Out*, 2004.